

Verify It Yourself — the IT Sheet

One page, both registers: the **four-click demo** for the business stakeholder, and the **independent-verification claims and tests** for the IT / SME reviewer.

For: the stakeholder carrying this into an IT / security review, and the SME receiving it. One page, both registers.

Read this first — the bridge. This is not new machinery. It is the exact trust system your IT team already operates — TLS certificates, SSH keys, signed OS updates — applied to business records: permissions, receipts, consent. There is no blockchain, no coin, no consensus protocol. Just digital signatures plus an append-only chain that anyone can recompute.

SIDE A — THE FOUR-CLICK DEMO (BUSINESS)

Open crm-sync.dev/verify and do this in front of the room:

1. **Say:** "This page checks a permission slip against our published public key, on your machine. It never asks the server whether to trust anything."
2. **Click "No token? Mint a sample."** — "The server just sealed a fresh permission slip — who may act, on what, until when — with a private key only the server holds."
3. **Click "Verify offline" → ✓ SIGNATURE VALID.** — "Your browser paired the published public half with that seal. Nobody vouched for it; the math did."
4. **Click "Flip one character" → Verify → ✗ fails.** — "One character changed after sealing, and it's caught. Records that cannot be quietly altered."

If anyone asks how, the "**How does this work?**" button on that page explains it in three steps (seal → pairing → ledger), with the everyday anchors: the SSH key

you add to GitHub, the chip in a bank card, the padlock behind `https://`, the `.env` rule — public half publishable anywhere, private half never leaves the server.

The sample token expires in 15 minutes — deliberately. Verify it again after coffee and watch it fail honestly: expiry is enforced by math too.

SIDE B — THE TECHNICAL CLAIMS (IT / SME)

Everything below is checkable without trusting the platform or its web page.

- **Format.** Tokens are compact JWS, signed EdDSA (Ed25519). The protected header names the signing key by `kid`.
- **Keys.** Public halves only, published at `https://crm-sync.dev/.well-known/jwks.json`. The private half stays server-side and is minted through a key ceremony. Rotation uses named generations: a new `kid` is published beside the old, retired keys keep verifying until their `not_after` — no flag day, no re-issuing history.
- **Independent verification.** Use your own tooling — not our page:

```
// Node - npm i jose

import { createRemoteJWKSet, jwtVerify } from "jose";

const JWKS = createRemoteJWKSet(new URL("https://crm-sync.dev/.well-known/jwks.json"));

const { payload, protectedHeader } = await jwtVerify(token, JWKS);

console.log(protectedHeader.kid, payload);
```

The hosted page at `/verify` is a WebCrypto convenience; its only network call is the JWKS fetch. Verification never contacts the platform API.

- **Possession is not authority.** The sample token carries `caps: []` — holding it grants nothing. Real tokens carry scoped capabilities that are checked

server-side on every call, and are revocable per credential (`jti`) and per subject, in real time. A leaked token is ejected without touching anyone else's access.

- **Audit.** Every certificate verification lands on a hash-chained register — `/license/verifications?jti=...` — timestamps, results, anonymized distinct-checker counts, and a `chain_intact` flag your auditor can recompute. History can be visibly broken, never quietly edited.

Tests to run

- ☐ Mint a sample at `/verify/sample` and verify it with your own JWKS tooling — not ours.
- ☐ Flip one byte anywhere in the token → verification must fail.
- ☐ Wait past the 15-minute expiry → verification must fail on `exp`.
- ☐ Fetch the JWKS and confirm it carries no private key material (OKP/Ed25519, `x` only).
- ☐ Ask for a rotation demonstration: after a rotation, old records still verify under the retired `kid`.

Endpoints

SURFACE	WHAT IT IS
<code>/verify</code>	Offline verifier — WebCrypto in your browser; mint-a-sample + tamper demo built in
<code>/verify/sample</code>	Mints a live, zero-authority demo token (15-minute expiry)
<code>/.well-known/jwks.json</code>	The published public key set
<code>/license/verify</code>	Server-side certificate check — appends to the chain of custody
<code>/license/verifications?jti=</code>	The per-certificate chain-of-custody register

This demo proves the mechanism with keys the platform holds. The follow-on conversation — when your IT says "fine, but we would hold our own keys" — is exactly the intended next step: tenant-scoped credentials and a signing generation of your own, minted under the same key ceremony, with separation of duties on your side. The demo earns that meeting; it does not preempt it.

Companion reading: [Your Firmware Is a URL — the CRA Assumes an Evidence Chain](#) · [Firmware, SBOM & the Cyber Resilience Act](#) · the deck [Entitlement for Composable AI \(PDF\)](#).
